10. The equation of a body in a rectilinear motion is $x = 3 - 6t + 3t^2$, where x is in meters, t is in seconds. At what time is the particle at rest.

Solution

The displacement of the rectilinear motion is $x = 3 - 6t + 3t^2$

Differentiate with respect to the time t, then

$$v(t) = \frac{dx(t)}{dt} = -6 + 6t$$

The particle is at rest when $v(t) = 0$

Hence,

$$0 = -6 + 6t$$
$$t = 1s$$

3.3 Freely Falling Bodies (Vertical motion under gravity)

An object falling under the influence of the earth's gravitational force or the motion of an object thrown vertically upward or freely falling downward is an example of uniform acceleration in a straight line. *All bodies at a particular location fall with the same downward acceleration, regardless of their size or weight.* The uniform acceleration of the object is called acceleration due to gravity when it is vertically downwards towards the center of the earth, it is denoted by **'g'** and it has an approximate value is **$9.8m/s^2$**.

Since the motion undergoing acceleration due to gravity is always downwards then **a = g**

And if the motion is vertically upwards then the acceleration is negative, **a = -g.**

The distance of a falling object is denoted by height (**h**).

The motion equations of falling bodies are the same as that of linear motion.

For vertically downward motion, **a = +g**

1. $v = u + gt$
2. $h = ut + \frac{1}{2} gt^2$
3. $V^2 = u^2 + 2gh$

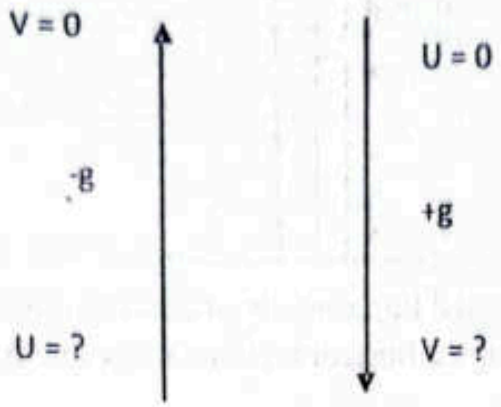
For vertically upward motion, **a = -g**

$v = u - gt$

$h = ut - \frac{1}{2} gt^2$

$V^2 = u^2 - 2gh$

Vertical motion diagram



3.3.1 Equation modifications for vertically upward motion

1. *Time taken to reach the maximum height*

$$V = u - gt = 0 \rightarrow t = \frac{u}{g}$$

2. *Maximum height*

Using $v^2 = u^2 - 2gh$, *sin* $v = 0$

i.e. $0 = u^2 - 2gh \quad h = \frac{u^2}{2g}$

3. *Time of flight*

This is the time taken for a body to travel from point of projection to the maximum point and back to the point of projection. i.e. the time of flight is twice the time taken to reach the maximum height.

$$t = \frac{2u}{g}$$

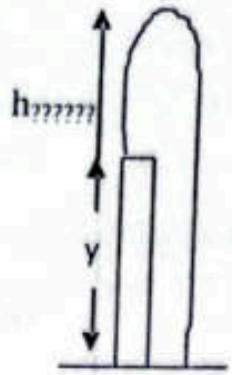
Special cases

1. If the particle is projected vertically at a point above the ground (i.e. at a distance y above the ground). then the maximum height reached from the ground is:

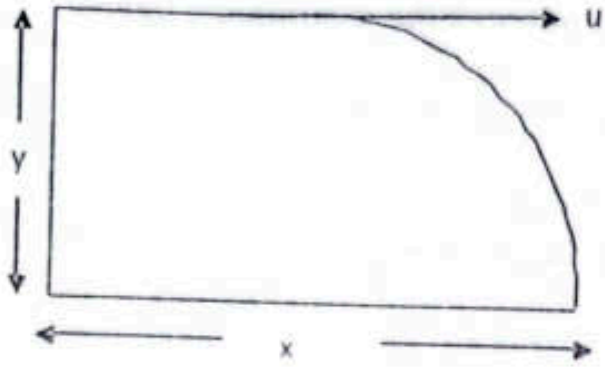
$$y + h_{max} = y + \frac{u^2}{2g}$$

The total distance travelled by the particle is:

$$y + 2h_{max} = y + \frac{u^2}{2g}$$



2. If the particle is projected horizontally at a point above the ground, the particle travels through a horizontal distance, x and a vertical distance, y



0)

The time taken to travel the horizontal distance x is the same as the vertical distance y.

Equation of motion for vertically downward motion.(since u – 0)

1. v = gt (From $v = u + at$)
2. $s = \frac{gt^2}{2}$ (From $s = ut + \frac{1}{2}at^2$)
3. $v^2 = 2gs$ (From $v^2 = u^2 + 2as$)

Solved Problems

1. A man throws a ball vertically upward with an initial speed 80m/s. wha the velocity of the ball on striking the ground?

Solution

at maximum height, $v = 0$,

using = u - gt

$$u - gt,\ t = \frac{u}{g} = \frac{80}{9.8} = 8.16s$$

$$v = gt = 9.8 \times 8.16 = 80m/s$$

2. A ball is thrown upwards on the ground with an initial velocity of 25m/s. Calculate

i) The time taken to reach the maximum height

ii) The maximum height reached by the ball

iii) Its velocity 2s after it has been thrown ($g = 9.8m/s^2$)

Solution

i) To calculate the time taken to reach the maximum height

$$v = u = gt = 0 \rightarrow t = \frac{u}{g}$$

$$t = \frac{25}{9.8} = 2.55s$$

It takes the ball 2.55s to reach the maximum height

ii) The maximum height reached by the ball can be obtain using:

$$h = ut - \frac{1}{2}gt^2$$

$$h = 25 \times 2.55 - \frac{1}{2} \times 9.8 \times 3.55^2$$

$$h = 31.89m$$

The maximum height reached is 31.89m

iii) For the velocity at t = 2s

$$v = u - gt$$

$$v = 25 - (9.8 \times 2) = 5.4m/s^2$$

3. A stone is thrown horizontally from the top of a tower 70m high with an initial velocity of 20m/s. How far from the tower does the ball hit the ground? ($g = 9.8m/s^2$)

Solution

The time taken to travel through the horizontal distance is the same as the time taken to travel through the vertical distance, considering the vertical position.

From

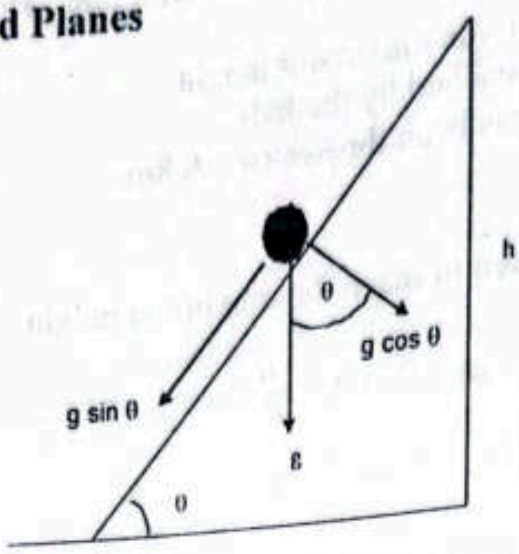
$$s = \frac{gt^2}{2}$$

$$t = \sqrt{\frac{2s}{g}} = \sqrt{\frac{2 \times 70}{9.8}} = 3.78s$$

Horizontal distance = 20 3.78 75.6 m

The ball hits the ground at a distance 75.6m from the tower.

3.4 Inclined Planes



The motion of an object freely rolling down an inclined plane elevated at an angle is a case of freely falling bodies, resolving along the plane gives $x = g \sin \theta$ and perpendicular to the plane is $y = g \cos \theta$

whichimplies that the effective part of the acceleration due to gravity used by the rolling object is divided into two components.

Motion of the object moving up the inclined plane

For upward motion, the acceleration is negative, therefore $= - g \sin \theta$

Equations of motion of the object up the inclined plane are:

1. $v = u - gt \sin \theta$
2. $s = ut - \frac{1}{2} (g \sin \theta) t^2$
3. $v^2 - u^2 - 2 (g \sin \theta)s$

Time taken to reach the maximum height ($v = 0$)

$$t - \frac{u}{g \sin \theta}$$

Maximum height up the inclined plane ($v - 0$)

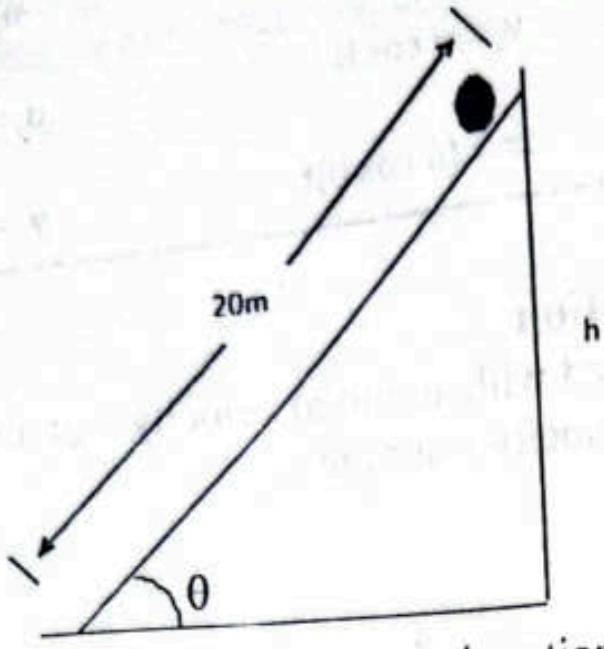
$$h_{max} = \frac{u^2}{2g \sin \theta}$$

Solved Problems

An inclined plane of inclination θ has a length of 20m. If a body rolls down the plane freely along the entire length with an acceleration of 7m/s². Calculate

height of the

n



ne object is moving on a downward motion. $a = +g\sin\theta$

$$\sin\theta = \frac{a}{g} = \frac{7}{9.8} = 0.71$$

$$\theta = \sin^{-1}(0.51) = 45.58^{0}$$

e angle of inclination of the plane is

om the diagram, using Pythagoras theorem.

$$\text{Sin}\,\theta = \frac{h}{20} \rightarrow \sin 45.58^{0} = \frac{h}{20}$$

$$h = 20 \times \sin 45.58^{0} = 14.28m$$

eight of the inclined plane is 14.28m

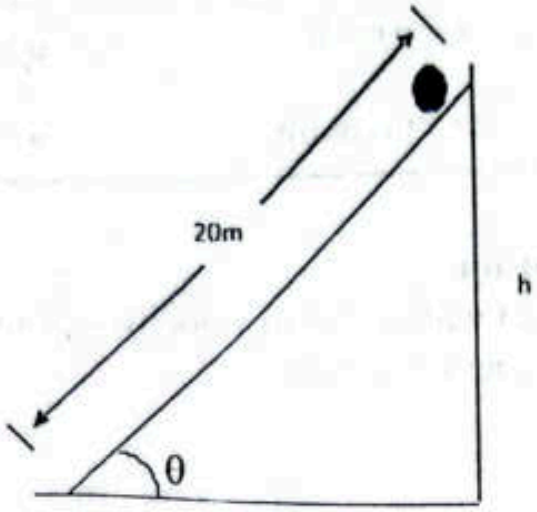
rojectile Motion

ectile is a body which is given an initial velocity and follows a path rmined by effects of gravitational acceleration and negligible air tance.

nitial velocity is vertical, then the particle will move in a straight line gravity, however, if its initial velocity is not vertical (inclined at an to the horizontal), then the particle will move in a circle (a parabola). ath followed by a projectile is called its **trajectory**. A projectile motion o-dimensional and we can treat the x and y coordinates separately. As n below

a) The angle of inclination θ
b) The height of the plane ($g = 9.8m/s^2$)

Solution



Since the object is moving on a downward motion. $a = + g \sin\theta$

$$\sin\theta = \frac{a}{g} = \frac{7}{9.8} = 0.71$$

$$\theta = \sin^{-1}(0.51) = 45.58^{\circ}$$

a) The angle of inclination of the plane is

b) From the diagram, using Pythagoras theorem.

$$\text{Sin}\,\theta = \frac{h}{20} \rightarrow \sin 45.58^{\circ} = \frac{h}{20}$$

$$h = 20 \times \sin 45.58^{\circ} = 14.28m$$

The height of the inclined plane is 14.28m

3.5 Projectile Motion

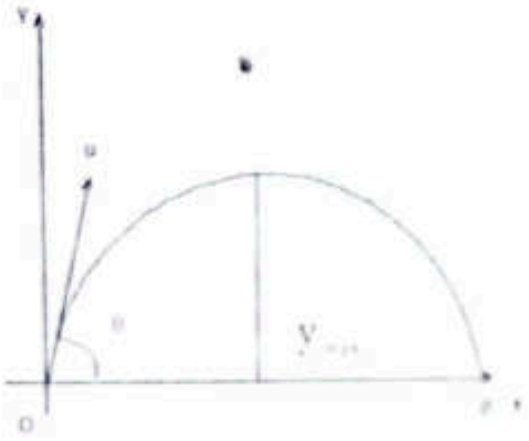
A projectile is a body which is given an initial velocity and follows a path determined by effects of gravitational acceleration and negligible air resistance.

If its initial velocity is vertical, then the particle will move in a straight line under gravity, however, if its initial velocity is not vertical (inclined at an angle to the horizontal), then the particle will move in a circle (a parabola). The path followed by a projectile is called its **trajectory**. A projectile motion is two-dimensional and we can treat the x and y coordinates separately. As shown below

	Horizontal motion	Vertical motion
Initial velocity	$u_x = u\cos\theta$	$u_y = u\sin\theta$
Final velocity	$v_x = u\cos\theta$	$u_y = \sin\theta - gt$
Distance travelled	$xr = (u\cos\theta)t$	$y = (u\sin\theta)t - \frac{1}{2}gt^2$

3.5.1 Motion Description

Consider a body projected with an initial velocity u at an angle to the horizontal and has a velocity v at time t.



The motion of the body projected horizontally and vertically should be analyse separately, then initial velocity u will be resolved into horizontal and vertical components as:

$u_x = u\cos\theta$ *and* $u_y = u\sin\theta$ respectively

3.5.2 Basic definitions

1. Time of flight

This is the time taken for the projectile to travel along its path from O to P

$$t = \frac{2u\sin\theta}{g}$$

2. Range

This is the distance OP travelled by the projected body.

$$R = \frac{u^2\sin 2\theta}{g}$$

At $\theta = 45$, the range is maximum, hence the maximum range

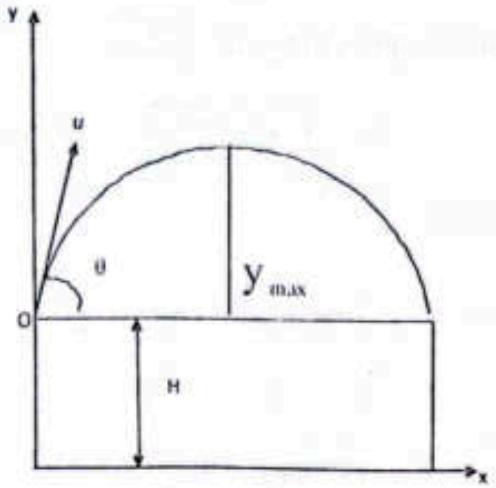
$$R_{max} = \frac{u^2}{g}$$

3. Maximum height

The projectile reaches its maximum height when the vertical velocity is zero.

$$Ymax = \frac{u^2 \sin^2 \theta}{2g}$$

If the body is projected from a point O above the ground, the initial height of the point O from the ground will be included.



Solved Problems

1. A particle is projected from the ground level with speed $40m/s^2$ at an angle of 60 to the horizontal. Calculate
   i) The time of flight
   ii) The range
   iii) The time taken to reach the greatest height
   iv) The greatest height [Take g = 10 m/s^2]

Solution

i). Time of flight, $t = \frac{2u \sin 0}{g}$

$$= \frac{2 \times 40 \times \sin 60^{\circ}}{10} = 7s$$

ii) The range. $R = \frac{u^2 \sin 2\theta}{g}$

$$= \frac{(40)^2 \sin 60^{\circ}}{10} = 138.6m$$

i) Time taken to reach the maximum height,

$$t = \frac{u \sin\theta}{g} = \frac{40 \sin 60^{\circ}}{10} = 3.5s$$

ii) Maximum height,

$$y_{max} = \frac{u^2 \sin^2 \theta}{2g}$$

$$= \frac{40^2 \sin^2 60^\circ}{2 \times 10}$$

$$= 60m$$

2. A parachutist after bailing out falls 80m without friction. When parachute opens, it decelerate it 6m/s^2. He reaches the ground with a speed of 4m/s. Calculate the bailing height.

Solution

Initial velocity can be determined using: $u = \sqrt{2gh}$

$$u = \sqrt{2 \times 80 \times 9.8} = 39.60/s$$

Final velocity is $v = 4m/s$

From $v^2 = u^2 + 2as$

$$4^2 = (39.60)^2 + 2 \times (-6) \times s$$

$39.60)2 + 2 \quad (-6) \quad s$

$$S = 129.35m$$

Total height = initial height + final height

$$= 80 + 129.35 \quad 209.35m$$

3. An aeroplane drops a parachutist. After covering a distance of 50m, he opens the parachute and retards at 5 ms^{-2}. If he reaches the ground with a speed of 3 m/s, what is the total time spent in the air?

Solution

The time t, taken before opening the parachute can be determine using:

$$h = ut + \frac{1}{2} gt_1^2$$

$$50 = 0 + \frac{1}{2} (9.8) t_1^2$$

$$t_1 = 3.19s$$

The velocity v_1 attained after falling through 50m is obtained using:

$$v_1^2 = u^2 + 2as$$

$$v_1^2 = 0 + 2(9.8)(50)$$

$$v_1 = 31.30m$$

To determine the time taken after opening the parachute. use

$$V = u + at_2$$

$$3 = 31.30 + (-5)t_2$$

$$t_2 = 5.66s$$

Total time t spent in air is there fore = $t_1 + t_2$

$T = 3.19s + 5.66s = 8.85s$

Practice Questions

1. A body changed velocity from 15m/s to 45m/s in 5s. Calculate the acceleration in metre per second square A.15 B. 45C. 6 D.5
1. An object is thrown vertically upwards from the floor with an initial velocity of 30m/s what is the total time spent by the object in the air?**(6sec)**
2. An apple fruit drops to the ground from the top of a tree 65m tall. How long does it take to reach the ground? ($g = 10m/s^2$). **(root 13)**
3. A body starts from rest and accelerates to a velocity of 15m/s in 20s. It maintains this velocity for 12s and then decelerates to rest in further 10s.
   i) Draw a velocity time graph of the motion
   ii) Calculate the initial acceleration of the body and the total distance covered by the body. **(405m)**
4. An elementary particle is projected in a dark hole, it moves spirally and its position is defined as $r - (t^2 + 7t)i - ej + 7 \sin 4tk$. Find its velocity and acceleration at $t = 0$, $t = 1$ and at what time is the particle at rest. **($v = 7i + 3j + 28k$, $v = 9i - 0.15j + 28k$), ($a = 7i - 9j$, $a = 7i - 0.45j - 7.8k$)**
5. The relation $3t = \sqrt{3x+6}$ describes the displacement of a particle in one direction. Where x is in meters and t is in seconds. Find the velocity of the particle at $t = 2s$. *($v = 24m/s$)*
6. A ball is projected horizontally from the top of a hill with a velocity of 20m/s. If it reaches the ground 4 s later, what is the height of the hill? **(ans: 80m)**
7. An object is projected with a velocity of 100m/s from the ground level at an angle θ to the vertical. If the total time of flight of the projectile is 10s, calculate the projected angle.**(ans: 29degrees)**
8. A stone is thrown horizontally with an initial velocity of 20m/s from the top of a building 80m high. Find the horizontal range of the stone. Calculate also the velocity with which the stone strikes the ground **($x=81m$, $v_y=39.7m/s$)**.
9. A ball was through from the ground and its velocity was 10m/s at an altitude of 15m above the ground. Calculate the velocity at the point of throw and the maximum height reached. **($v = 10m/s$, $h = 5.1m$)**
10. A body is thrown forward from the top of a cliff 20m high with a horizontal velocity of 15m/s. calculate.

i. The time to reach the ground **($t = 2s$)**

ii. The horizontal distance from the foot of a cliff at which the object strikes the ground **($h=30m$)**.

11. An object is launched at a velocity of 40m/s in a direction making an Angle of 50° upward with the horizontal. The maximum height and the horizontal range of the object are ..... and ..... respectively. A. 6.25m & 47.9m **B. 47.9m and 161m** C. 6.25km & 47.9km D. 47.9km and 6.25km

12. An object is launched at a velocity of 20m/s in a direction making an angle of 30 degrees upward with the horizontal. Determine the time of flight [take g=10m/s^2] **A. 2.00seconds** B. 40minutes C. 4.0seconds D. 35metres

13. A body changed velocity from 15m/s to 45m/s in 5s. Calculate the acceleration in metre per second square A. 15 B. 45 **C. 6** D. 5

14. A body with initial velocity 135m/s was projected at angle 45 degrees What is the time taken to reach the maximum height? **A. 11.2s** B. 9.74s C. 9.81s D. 6.78s

15 A parachutist after bailing out falls 50m without friction. When parachute opens, it decelerate it 2m/s^2. He reaches the ground with a speed of 3m/s. At what height, did he bail out nearly. (**Ans: 298m**)

16 An aeroplane drops a parachutist. After covering a distance of 40m, he opens the parachute and retards at 2 ms^{-2}. If he reaches the ground with a speed of 2 m/s, what is the total time spent in the air? {**Ans: 16s**}

CHAPTER FOUR

DYNAMICS

Having discussed kinematics (in chapter three), which was the description of motion without considering what causes the motion, it is evident we need to know what causes the motion that is, the dynamics of motion and its changes. Force causes motion, so the study of dynamics is the description of motion with respect to force.

A force can be a push or a pull and it produces a change in the velocity of the object on which it acts

There are two types of force, **contact** and **non-contact**

(a) **Contact force** can be defined as any kind of forces that requires being in contact with another object.

Examples of contact force are:

Tension force: force applied by a full stretched cable

Normal force: opposing force on a stationary object that acts in an opposite direction to the pulling force of gravity

Frictional force: resisting force that exists when an object is moved or tries to move on a surface

Spring force: force exerted by a compressed or stretched spring.

Air resisting force: frictional force experience by an object moving through the air.

(b) **Non-contactforce** is the force that does not require contact with the object.

Examples of non-contact forces:

Electrostatic force: forces exerted by all electrically charged bodies on another charged bodies. These forces can be repulsive or attractive in nature based on the charge of the bodies.

Magnetic Force: force exerted by a magnet on the surrounding objects

Gravitational force: an attractive force that pulls objects with mass toward each other.

4.1 NEWTONS LAWS OF MOTION

Newton's First Law of Motion (the law of inertia)

Newton's first law states that if a body is at rest or moving at a constant speed in a straight line, it will remain at rest or keep moving in a straight line at constant speed unless it is acted upon by a force

According to Newton, the concept of inertia can be related to mass. All matter has a built-in reluctance to being moved if it is at rest, or having its motion changed if it is moving, mass is a measure of inertia of a body, that is, huge

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According to Newton, the concept of inertia can be related to mass. All matter has a built-in reluctance to being moved if it is at rest, or having its motion changed if it is moving, mass is a measure of inertia of a body, that is, huge

objects has more inertia or more reluctance to change its motion than lighter objects.

The inertia of a body is its tendency to remain at rest or continue its motion in a straight line unless it is acted upon by an external force to change its state

Newton's first law and the concept of inertia are identical; hence the law is sometimes called
the law of inertia.

Linear momentum

The momentum of a body is the product of its mass and velocity when it is in motion.

Momentum $(p) = mass \times velocity = mv$

S.I. unit of momentum is kgm/s or Ns. Momentum is a vector quantity

Newton's Second Law of Motion

It states that the time rate of change of momentum of an object is directly proportional to the force acting on it and takes place in the direction of the force.

Mathematically the second law is expressed as

$$\text{Force (F) A} = \frac{\text{Change in momentum } (mv - mu)}{\text{time taken for the change } (t)}$$

$$F = \frac{mv - mu}{t}$$

.1.1 Relationship between Force, Mass and Acceleration

$$F = \frac{mv - mu}{t} \rightarrow F = \frac{M(v - u)}{t}$$

$$\text{Recall that } \frac{v - u}{t} = a \rightarrow F = ma$$

Where k is the constant and k = 1, then

'F = ma'

Force can therefore be defined as the product of mass and acceleration

$Force = mass \quad acceleration$ i.e. $F = ma$

Suppose a constant Force is applied at different times to two bodies of

masses m_1 and m_2 and producing acceleration a_1 and a_2 respectively, then

$$\frac{m_1}{m_2} = \frac{a_2}{a_1}$$

The greater the mass of the body, the greater the force required for a constant acceleration and the greater the force the greater the acceleration for a fixed mass.

The impulse of a force is equal to the change in momentum that is

$$I = Ft = mv - mu$$

The impulse of a force can also be defined as product of force and time, S.I. unit is Ns

Solved Examples

1. A force of 10N acts on a mass of 5kg in 15s. Find the change in momentum and the acceleration of the body.

Solution

Since change in momentum is the same as the impulse of the force, then

$$I = Ft = mv - mu$$
$$I = 10 \times 150Ns$$

To calculate the acceleration, recall that

$$F = ma$$

$$a = \frac{F}{m} = \frac{10}{5} = 2m/s^2$$

2. An object in a trolley of total mass of 50kg has a velocity of 5 m/s. The trolley moves faster than expected for 8s and the velocity increases to 9 m/s. Calculate the force produced.

Solution

From Newton's second law

$$F = \frac{mv - mu}{t}$$

$m = 50kg,\ u = 5m/s, v = 9m/s,$ and $t = 8s$

$$F = \frac{50(9 - 5)}{8} = 25N$$

3. A car with a mass of 1200kg accelerates from 5m/s to 15m/s. The force of the engine acting on the car is 6000N. Over what distance did the force act.

Solution

Apply work kinetic theorem. it state that the work done on the body is equal to change in kinetic energy of the body i.e. F.d. $= \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

Then, 600 d $= \frac{1}{2}(1200)(15)^2 - \frac{1}{2}(1200)(5)^2$

$d = 20m$

Newton's Third Law

"It is states that to every action there is an equal and opposite reaction"

This law implies that if a body A exerts a force (action) on a body B, then B will exert an equal and opposite force (reaction) on A. Mathematical newton's third law can be expressed as:

$$F_{AB} = F_{BA}$$

4.2 The Law of Conservation of Linear Momentum

"It states that provided no external force acts on a system of colliding bodies then sum of momentum before collision is equal to the sum of momentum after collision"

Consider two bodies A and B of masses m_1 and m_2 respectively moving along the same straight line. If u_1 and v_1 represent initial and final velocity of A and u_2 and v_2 the initial and final velocity of B. then by the principle of conservation of linear momentum.

Total momentum before collision = Total momentum after collision

From Newton's second law.

$$Ft = mv - mu$$

If there is no external force acting $F = 0$

$$0 = mu - mu, \quad mv = mu$$

There are two types of collisions. These are:

1. Elastic collision
2. Inelastic collision

4.2.1 Elastic Collision

A collision is said to be elastic when both momentum and kinetic energy are conserved in the event of collision.

The equations describing the type of collision are

1. $M_1u_1 + m_2u_2 = m_1v_1 = m_2v_2$